PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience in natural language processing (NLP) and document classification. Experienced in taking machine learning work from research and prototyping through production deployment, including model serving and production quality monitoring. PhD candidate researching Croatian-language processing, with a focus on named entity recognition in legal texts.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer *03/2023 – Present*
Slavonica Document AI d.o.o., Osijek

- Developed an automated classification model for incoming documents, including invoices, contracts and administrative decisions, achieving 94% accuracy across 23 classes for public-sector clients.
- Fine-tuned BERTić and XLM-R transformer models for named entity recognition on Croatian documents.
- Introduced MLflow for experiment tracking and model registry; deployed models as FastAPI services in Docker containers on Azure ML and Azure App Service.
- Established production model quality monitoring for data drift, with weekly reporting.
- Mentor two students during professional internships.

Data Scientist *09/2021 – 02/2023*
Baranjska Kreditna Banka d.d., Zagreb (remote from Osijek)

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Analysed customer complaint text and built topic classification models using TF-IDF with logistic regression, later using fastText.
- Produced SQL reports using the Oracle data warehouse.

Research Project Assistant *10/2020 – 08/2021*
Faculty of Applied Informatics, Drava University

- Built a corpus of Croatian legal texts and annotated the data.
- Wrote Python scripts to support research work.

EDUCATION

PhD studies in Computer Science *2022 – Present*
Drava University
Research topic: Named entity recognition in Croatian legal texts

MSc in Computer Science *2020*
Drava University

TECHNICAL SKILLS

Python, PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, pandas, NumPy, SQL, MLflow, FastAPI, Docker, Azure ML, Git, Linux

LANGUAGES

Croatian (native), English (C1), German (B2)